

Article

Optimizing Renewable Energy Systems Placement Through Advanced Deep Learning and Evolutionary Algorithms

Konstantinos Stergiou¹ and Theodoros Karakasidis^{2,*} ¹ Civil Engineering Department, University of Thessaly, 38334 Volos, Greece; kostergiou@uth.gr² Physics Department, University of Thessaly, 35100 Lamia, Greece

* Correspondence: thkarak@uth.gr

Abstract: As the world shifts towards a low-carbon economy, the strategic deployment of renewable energy sources (RESs) is critical for maximizing energy output and ensuring sustainability. This study introduces GREENIA, a novel artificial intelligence (AI)-powered framework for optimizing RES placement that holistically integrates machine learning (gated recurrent unit neural networks with swish activation functions and attention layers), evolutionary optimization algorithms (Jaya), and Shapley additive explanations (SHAPs). A key innovation of GREENIA is its ability to provide natural language explanations (NLEs), enabling transparent and interpretable insights for both technical and non-technical stakeholders. Applied in Greece, the framework addresses the challenges posed by the interplay of meteorological factors from 10 different meteorological stations across the country. Validation against real-world data demonstrates improved prediction accuracy using metrics like root mean squared error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE). SHAP analysis enhances transparency by identifying key meteorological influences, such as temperature and humidity, while NLE translates these insights into actionable recommendations in natural language, improving accessibility for energy planners and policymakers. The resulting strategic plan offers precise, intelligent, and interpretable recommendations for deploying RES technologies, ensuring maximum efficiency and sustainability. This approach not only advances renewable energy optimization but also equips stakeholders with practical tools for guiding the strategic deployment of RES across diverse regions, contributing to sustainable energy management.



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1. Introduction

The European Union (EU) has set ambitious goals to reduce its dependency on fossil fuels as part of its broader strategy to combat climate change. These goals have been supported by a range of policies and regulations, including a cap-and-trade system designed to limit carbon dioxide emissions from power plants and other major sources [1]. Such initiatives are crucial components of the EU's drive toward a more sustainable and low-carbon economy. Within this context, renewable energy sources like solar and wind power are gaining prominence as vital, eco-friendly methods for energy production. However, the efficiency of these renewable sources is heavily influenced by climatic factors, which introduces significant challenges in predicting energy production.

RES, such as solar and wind, are critical components in the fight against climate change and the transition to a low-carbon economy. However, the strategic deployment and optimization of these energy sources present several challenges due to the inherent variability of renewable resources. The efficiency of renewable energy systems is highly dependent on environmental factors, such as sunlight, wind speed, and temperature, which can vary significantly across different geographical locations and over time. This variability

adds complexity to the tasks of predicting energy output and efficiently placing renewable energy systems to achieve maximum energy production.

To overcome these challenges, the integration of advanced predictive models and optimization techniques is essential. Traditional energy planning methods are often limited in their ability to handle the spatial and temporal complexities of renewable energy. As a result, there is a growing need for sophisticated, data-driven frameworks that can effectively analyze large volumes of meteorological data, optimize the placement of RES, and make informed decisions that enhance both sustainability and energy output. Machine learning (ML) and AI have emerged as powerful tools in this regard, offering the potential to analyze complex datasets and provide insights that can significantly improve the strategic planning and deployment of renewable energy technologies.

2. Literature Review

ML algorithms have emerged as powerful tools in many domains, from load forecasting [2] and enabling the analysis of meteorological data to forecasting the output of RESs such as solar farms and wind turbines [3,4]. By utilizing advanced predictive models and ML techniques, these algorithms can significantly enhance the accuracy and reliability of renewable energy predictions, thereby optimizing the placement and operation of RES. Moreover, ML can detect patterns in meteorological data, further boosting the overall performance and efficiency of renewable energy systems. As a result, ML plays a critical role in the ongoing transition to a more sustainable and efficient energy system. Recent studies have underscored the value of integrating ML with deep learning models to improve renewable energy forecasting. For instance, Al Kandari and Ahmad [5] proposed an ensemble approach that combines deep learning models such as long short-term memory (LSTM) and gated recurrent unit (GRU) with statistical models, achieving notable improvements in forecasting accuracy. Similarly, Chang et al. [6] have addressed the challenges of integrating solar power into electrical systems by introducing the TESDL approach, which utilizes deep learning to predict weather patterns, thereby enhancing prediction accuracy by 27% compared with traditional methods. Research has also demonstrated the benefits of using convolutional neural networks (CNNs) to capture complex weather patterns. Zhang et al. [7] reduced forecast error rates by employing deep CNNs, showcasing the potential of these models to enhance solar farm forecasting accuracy. Jebli et al. [8] explored the application of ML and deep learning in estimating solar energy production, finding that models such as random forest (RF) and artificial neural networks (ANNs) offer superior precision for real-time and short-term forecasting compared with traditional models.

Recent advancements in hybrid deep learning models, such as those found in the work of Khan et al. [9] and Rai et al. [10], highlight the potential of combining different deep learning architectures for improved prediction stability and accuracy. These approaches leverage the strengths of individual models, such as ANN and GRU, to create robust forecasting systems that can adapt to varying meteorological conditions. Beyond solar energy, wind energy forecasting has also benefited from the application of ML and deep learning techniques. Agga et al., 2022 [11] have presented a deep learning model that merges the LSTM and convolutional neural network (CNN) architectures for the accurate prediction of solar power production. The model aimed to improve the operation of solar power systems by providing more precise estimates of power generation, which is essential for delivering prime energy to consumers and ensuring the reliability of power plants' operations. Using real-world data from Rabat, Morocco, the model was analyzed, and it was determined that it outperformed traditional machine learning and single DL models in terms of prediction accuracy, precision, and stability as measured by error metrics such as MAE, MAPE, and RMSE. Pombo et al., 2022 [12], introduced a general physical model of a photovoltaic (PV) system with ML predictions for one- to three-day forecasting. The operating point of the PV is captured by the model using measurements of power, wind speed, and air temperature, which are recombined into physics-informed metrics. This enabled the models to understand the physical correlations between the many

characteristics, hence simplifying the training process. The case study of a PV system in Denmark was utilized to test the claims by assessing the following five alternative ML techniques: RF, support vector machines (SVM), CNN, LSTM, and a CNN–LSTM hybrid. The findings demonstrate that the top predictors constantly use the recommended physics-based characteristics, and that the forecasting horizon has no effect on the outcomes. The proposed method achieved an RMSE of 7.5%.

Shabbir et al., 2019 [13] presented an SVM-based regression algorithm that was utilized to estimate wind energy output in Estonia one day in advance. The suggested method was afterwards compared with the prediction algorithm used by the Estonian energy regulating agency. Results reveal that the proposed approach offered more accurate forecasts and the lowest RMSE values. The research emphasized the application of an SVM-based regression method for predicting wind energy output, which can enhance supply and demand balance in power networks. Based on a high-frequency SCADA database with a 1-s sampling rate, Lin and Liu 2020 [14] developed a deep learning neural network to forecast wind power. On the basis of the physical process of offshore wind turbines, the linear and nonlinear correlations of the input characteristics were explored. Eleven factors, including wind speeds at various heights, pitch angles of each blade, average blade pitch angle, nacelle orientation, yaw error, and ambient temperature, were included in the prediction model. With an R^2 of 0.91, the findings demonstrate that the suggested method may minimize computing cost and time while retaining high accuracy in wind power forecasting. Demolli et al., 2019 [15] applied five different ML algorithms to forecast wind power values. Specifically, Lasso regression, kNN regression, XGBoost regression, RF regression, and support vector regression (SVR) have been implemented and the dataset that they utilized comprises 5 years of hourly wind speed observation values from Nigde, Turkey. SVR and RF outperformed the other ML models by achieving an RMSE of 93.13 and 123.88, and an MAE of 32.63 and 42.76, respectively. Moreover, Khan et al., 2019 [16] proposed a hybrid approach which combined principal component analysis (PCA) and deep learning to forecast wind power. PCA was used to extract hidden characteristics from wind data and eliminate strong correlations between values, while a TensorFlow-optimized deep learning system was employed to anticipate wind power based on relevant characteristics. The suggested technique was evaluated on three distinct datasets (hourly, monthly, and annually) from the National Renewable Energy Laboratory (NREL), and the findings were found to demonstrate that it successfully predicted wind output throughout a variety of time scales, from hours to years. Some of the existing applications of ML for wind power forecasts are based on big data analytics [17], the importance of atmospheric turbulence and stability [18], and interpretable physics [19].

On the other hand, evolutionary algorithms are a type of optimization technique based on the principles of natural selection. In the context of renewable energy production, evolutionary algorithms can be employed to improve ML models' accuracy or to determine the layout of a renewable energy system that maximizes production and minimizes costs. In addition, evolutionary algorithms can be used to optimize the functioning of renewable energy systems in real time, responding to changing weather conditions and other variables that might have an influence on the generation of renewable energy. Zhang et al., 2019 [20] presented a novel hybrid optimization algorithm for the optimal sizing of a stand-alone hybrid solar and wind energy system in remote and island areas where weather data are often unavailable. Their algorithm is based on three optimization techniques: chaotic search, harmony search, and simulated annealing. Weather forecasting data were utilized along with artificial neural networks to improve the accuracy of the results, and the objective function focused on minimizing the total life cycle cost while accounting for system reliability. Their results show that the proposed hybrid optimization algorithm with weather forecasting data could improve the efficiency and feasibility of a stand-alone hybrid renewable energy system for remote and island areas. For the prediction of solar power production, Li et al., 2019 [21] suggested a novel hybrid optimization technique termed hybrid improved multi-verse optimizer (HIMVO). Their HIMVO algorithm was

used to optimize the support vector machine (SVM) for photovoltaic output prediction by introducing chaotic sequences to initialize the population, which significantly improved the algorithm's convergence rate in comparison with other optimization algorithms such as the multi-universe optimizer algorithm, particle swarm optimization algorithm, and dragonfly algorithm. Their results show that the HIMVO method had higher optimization capability and stability, and that the HIMVO–SVM model had superior prediction accuracy and stability compared with other models. Eseye et al., 2018 [22] introduced a hybrid forecasting model, hybrid WT–PSO–SVM, for short-term (one day in advance) production power forecasting of a real microgrid PV system. Using wavelet transform, particle swarm optimization, and support vector machine, the model enhanced the accuracy of forecasts. Using interactions between PV system supervisory control and data acquisition (SCADA) real power records and numerical weather prediction (NWP) meteorological data, the model adjusted the SVM parameters using particle swarm optimization (PSO) to improve forecasting precision. In comparison with seven previous forecasting methodologies, the suggested model demonstrated superior predicting accuracy. Mean absolute percentage error (MAPE) is 4.2% on a daily basis, whereas normalized mean absolute error (NMAE) is 0.4%.

Moreover, incorporating XAI into the GREENIA framework allows for greater transparency and understanding of how various environmental factors influence energy production predictions. The use of Shapley additive explanation (SHAP) values in GREENIA exemplifies this approach, providing insights into the contributions of different variables such as wind speed and temperature, towards the model's predictions [23]. This transparency is vital for building trust among stakeholders, including policymakers and investors, who require assurance that AI-driven decisions are both informed and justifiable [24]. The positioning of RES is essential for maximizing energy output and improving grid dependability. Diverse decision-making systems have been established to enhance this process, including geographic information systems (GISs) and optimization techniques. Geographic decision support systems (GDSSs) utilize GISs to identify optimal locations for distributed energy resources (DERs). This approach has aided utility companies in making informed decisions regarding DER portfolios, as demonstrated in Claremont, California [25]. More particularly, in Greece, a comprehensive siting procedure was developed using GISs to address the challenges of RES placement. This method considered multiple parameters affecting site selection, exemplified by a case study on the island of Syros [26]. While existing frameworks, such as GDSS and optimization techniques like unified particle swarm optimization (UPSO) [27], have been instrumental in improving RES placement, the GREENIA framework introduces a new level of sophistication. By integrating advanced ML models, such as GRUs, with evolutionary algorithms like the Jaya optimization technique, GREENIA not only predicts energy output with accuracy but also optimizes RES placement based on dynamic meteorological and geographical data. Furthermore, the use of XAI techniques within GREENIA ensures that the decision-making process is transparent, providing insights into the key factors influencing energy production. This level of interpretability, combined with prediction and optimization capabilities, makes GREENIA a powerful tool for policymakers and energy planners, driving more effective, data-driven decisions for renewable energy deployment. For instance, the island of Crete can serve as a prime example of how local communities are advancing towards energy independence using diverse renewable resources. The development of community energy labs (CELs) across the island, as part of the 'Renewable Energy Valleys' (REVs) initiative, aims to fully meet local energy demands through renewable energy sources such as solar, wind, geothermal, and biomass. This initiative is not only a demonstration of a data-driven multi-carrier grid management system but also an indicator of the renewable energy potential in Greece [28].

Classic optimization algorithms, while foundational in various fields, exhibit several notable drawbacks that limit their effectiveness in complex problem-solving scenarios, such as repetition in the local minimum [29]. These limitations stem from the reliance of these algorithms on specific assumptions and conditions, which can lead to suboptimal results

when applied to diverse or dynamic problems. The GREENIA framework as a holistic integration provides several advantages over existing methods. Firstly, GRUs effectively model temporal relationships, making them well suited for forecasting energy production based on historical meteorological data, which is essential for accurately predicting renewable energy output. Additionally, we use the Jaya evolutionary algorithm for optimizing the placement of RES. Unlike traditional gradient-based methods, Jaya can efficiently find global optima without becoming stuck in local optima, which is particularly beneficial for the complex, non-convex optimization landscapes typical in renewable energy scenarios. Finally, classical methods often lack transparency for stakeholders without technical expertise. The proposed framework, however, integrates SHAP and NLE, which provide interpretability and actionable insights, ensuring that the model's outputs are accessible and understandable for both technical and non-technical stakeholders.

In summary, GREENIA transcends traditional methods by offering an integrated, AI-driven, and explainable framework for optimizing the placement of renewable energy sources (RES) across varied geographic settings. It utilizes evolutionary algorithms to determine optimal RES placement, specifically applied to a case study in Greece. The study is organized as follows: Section 1 introduces the problem, laying out the challenges in RES deployment. Section 2 reviews the existing literature and methods currently employed. Section 3 details the methodology underpinning the GREENIA implementation, while Section 4 presents the results of applying the GREENIA approach, comparing them to other state-of-the-art models. Section 5 provides a comparative analysis between the outcomes of the proposed framework and those from existing literature. Section 6 discusses the limitations of the approach and future research directions, and, finally, Section 7 offers a comprehensive conclusion summarizing the implementation and the overall contribution of the GREENIA framework.

3. Methodology and Models

This section provides detailed information about the GREENIA methodology, with an emphasis on the models and metrics employed within the framework. The high-level methodology of the GREENIA framework, as illustrated in Figure 1, encompasses a comprehensive pipeline for RES placement optimization. The figure provides a visual representation of the interconnected processes, which include data preprocessing, ML-based energy prediction, optimization, explicability, and the generation of natural language explanations for actionable insights.

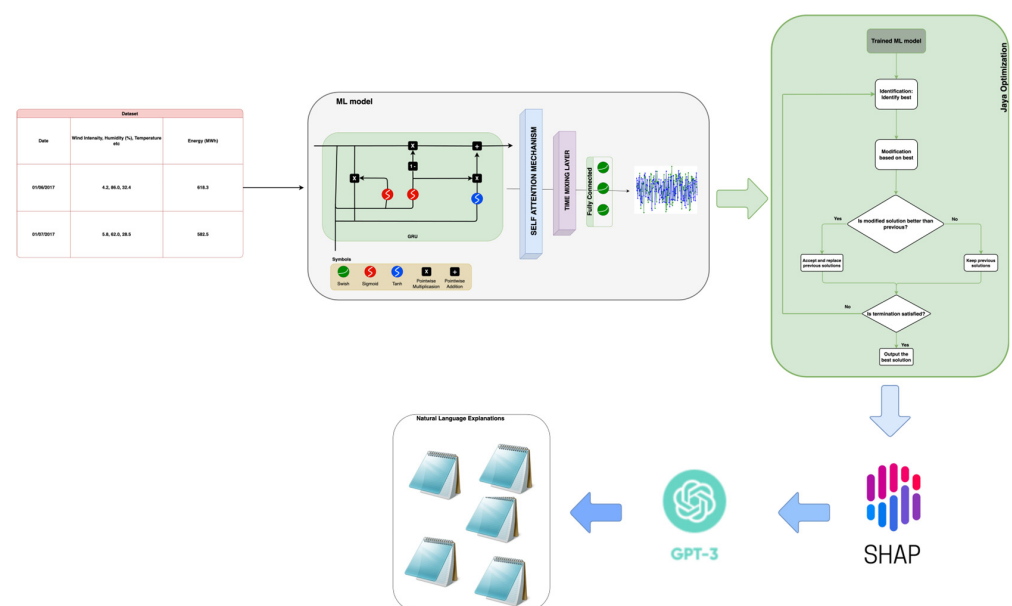


Figure 1. GREENIA high-level architecture.

The framework begins with the collection and preprocessing of meteorological and energy production data. This includes features such as wind intensity, humidity, temperature, and energy production (in MWh), ensuring high-quality inputs for the predictive model. The core predictive model is a GRU-based neural network enhanced with attention mechanisms and time mixing layers. The GRU layer captures temporal dependencies within the sequential data, while the attention mechanism emphasizes critical features or time steps. A series of fully connected layers with Swish activation transforms the outputs, leading to precise energy production forecasts. The trained ML model outputs are optimized using the Jaya algorithm. The optimization module refines the input solution iteratively to maximize RES production. The flowchart includes decision nodes for evaluating and improving solutions based on predefined criteria. To ensure transparency, Shapley additive explanations (SHAPs) were applied to analyze the influence of input features (e.g., temperature, humidity) on model predictions. This provided stakeholders with an interpretable understanding of the forecasting process. Outputs from the SHAP analysis were translated into user-friendly, human-readable recommendations using GPT-3. These explanations enhanced accessibility for non-technical stakeholders, such as policymakers and energy planners. The framework culminated in actionable insights and strategic recommendations for deploying RES technologies, ensuring maximum efficiency and sustainability.

3.1. Data Collection and Preprocessing

3.1.1. Data Sources

The primary datasets used in this study were collected from meteorological stations (<http://www.emy.gr/emyl/el/>) accessed on 20 May 2021, and located across various regions of Greece, including Agchialos, Aktio, Alexandroupoli, Araxos, Argos, Florina, Ioannina, Kalamata, Kavala, and Kozani. These datasets include key meteorological parameters—such as temperature, humidity, wind speed, and solar radiation—that are crucial for predicting renewable energy production. Additionally, historical energy production data (<https://www.admie.gr/>) accessed on 20 May 2021, measured in megawatt-hours (MWh), were obtained from regional energy authorities to serve as the target variable in the predictive modeling process.

In Figure 2, Greece's map, along with the corresponding meteorological stations' locations, is presented. The map illustrates the whereabouts of the meteorological stations utilized for collecting input data for the analysis of renewable energy implementation in Greece. These stations supplied critical data on the wind speed, wind direction, temperature, and humidity, which are crucial for enhancing the positioning of RES. Furthermore, these sites correspond with the regions for which the strategic plan and placement recommendations were produced, based on the analysis performed utilizing the GREENIA framework. The green markers denote the locations of meteorological stations in areas including Kozani, Alexandroupoli, Nea Agxialos, Ioannina, Araxos, Aktio, Kalamata, Argos, Kavala, and Florina. The stations supplied the meteorological data utilized to evaluate the renewable energy potential in each region. The areas identified by the meteorological stations are the same regions from which strategic recommendations for renewable energy deployment have been derived. The strategic plan provides recommendations for the optimal placement of wind turbines, solar panels, and other renewable energy system technologies, based on the data collected from these areas. The map's terrain colors emphasize the elevation and geographical characteristics of the areas. Darker regions indicate elevated terrain (such as hilly locations in Florina and Kozani), which may be more conducive to wind energy initiatives. Brighter regions, such as Kalamata and Nea Agxialos, indicate lower altitudes, rendering them more conducive for solar energy implementation. Through the analysis of meteorological data from strategically positioned stations, the GREENIA framework generated site-specific recommendations for the deployment of renewable energy sources, optimizing energy production while addressing sustainability and community requirements.

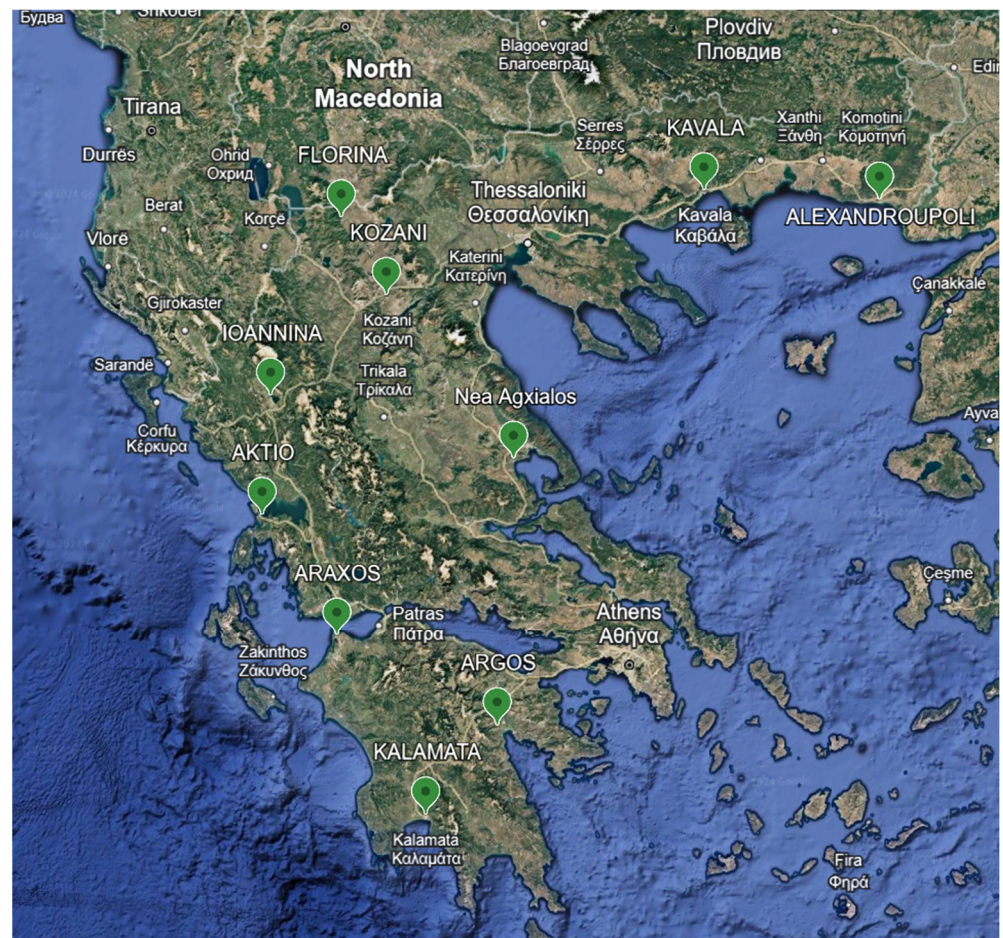


Figure 2. Map of meteorological stations across Greece.

3.1.2. Data Preprocessing

To ensure the consistency and usability of the data for machine learning purposes, several preprocessing steps were undertaken:

- **Column Renaming:** Each dataset was originally named based on the station's reporting structure, which varied across locations. To harmonize the datasets, column names were standardized by appending the station name to each feature (e.g., 'Temperature_agxialos').
- **Index Alignment:** Time-based data from different stations had to be synchronized. This was done by unifying the date and time fields across all datasets, converting them into a single 'Date' index. This enabled the seamless merging of datasets from different stations.
- **Data Resampling:** Since the data were recorded at different intervals (hourly, daily, etc.), it was resampled to a daily frequency. This resampling was essential to align the meteorological data with the daily energy production data.
- **Data Merging:** The meteorological data from different stations were merged into a single dataset using the 'Date' index as a common key. This comprehensive dataset included all relevant features required for energy production modeling.
- **Missing Data Handling:** Missing values were handled in a two-step process:
 - Columns with more than 65 missing values, which correspond to approximately 3.4% of the total 1900 time points (65/1900), were dropped to prevent excessive noise in the dataset and to maintain the integrity of the time series analysis.
 - For the remaining missing values, imputation was performed using the mean of the nearest non-missing values within the same time window. This method

preserves the local data structure and continuity, minimizing any potential bias introduced by gaps in the time series data, and ensuring that the overall temporal patterns remain intact.

- **Data Dimension:** The dataset used in this study consisted of daily meteorological data from 10 weather stations over 4 years (2015–2018). This resulted in the following:
 - **Number of records (rows):** 1009 (after preprocessing, accounting for missing or incomplete data).
 - **Number of features (columns):** Multiple meteorological variables, including wind intensity, humidity, temperature, and solar radiation. The number of the dataset's features was 49.

3.1.3. Data Cleaning

Additional data cleaning was necessary to ensure that all features were in the correct format for modeling. Non-numeric columns were removed, and all remaining features were converted to a consistent numeric format. This step was crucial for the success of subsequent machine learning algorithms, which require numeric input data.

3.2. Feature Engineering

3.2.1. Feature Scaling

To ensure that no single feature dominated the model due to differences in scale, specific scaling techniques to the datasets were applied. The input features (X), represented in the histogram in Figure 3, were scaled using the StandardScaler, which transforms the data so as to have a mean of 0 and a standard deviation of 1. The rescaling resulted in data within an approximate range of $[-2.5, 2.5]$, as visualized in the histogram, helping to improve the model's training and convergence. For the target variable (y), shown in Figure 4, the MinMaxScaler has been applied, which scaled the energy production values to a range of $[0, 1]$. The histogram of the scaled target values highlights this transformation, ensuring that the model's output was consistent with the scaled input features.

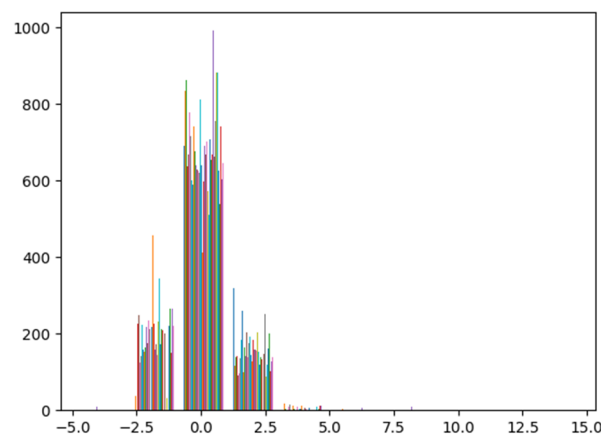


Figure 3. Histogram of the input features dataset. (Different colors correspond to different features).

The histogram in Figure 3 showcases the distribution of the input features after applying the StandardScaler. As seen in the figure, the majority of the data points are clustered around 0, with a significant concentration within the range of -2.5 to 2.5 . This tight clustering around the mean of 0 indicates that the standardization process was successful in centering the data. A small number of outliers are visible beyond this range, suggesting that, while most of the features fall within a normalized range, there are some values that exhibit more variance. This distribution highlights how the StandardScaler equalizes the spread of features, reducing the risk of any single feature disproportionately influencing the model's learning process. The multiple colors in the input features histogram correspond

with the various features in the dataset, explaining the presence of different distributions within the same figure.

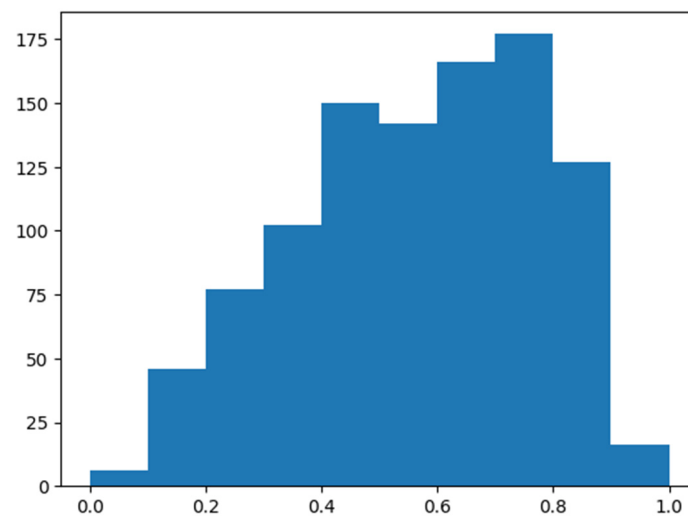


Figure 4. Histogram for the output feature.

The histogram in Figure 4 represents the distribution of the target variable (energy production) after being scaled using the MinMaxScaler. As expected from this scaling method, the data are rescaled to fall within the range of $[0, 1]$. The distribution shows that the majority of the target values are evenly spread across this range, with a slight concentration towards the middle (around 0.5 to 0.8), indicating that energy production values are neither too low nor too high in most instances. This transformation ensures that the target values were on a comparable scale with the input features, facilitating smoother learning and improved model performance.

In general, the x axes in the two histograms depict the normalized values of the input X and target y subsequent to the implementation of the corresponding scaling methods. The y axes represent the frequency of occurrences for each scaled value in the dataset, illustrating the distribution of the data post-scaling.

3.2.2. Sequence Padding

To handle the time-series nature of the data, the processed features were organized into sequences of 10 time-lag steps, allowing the model to learn temporal dependencies within the data. Different time-step lengths were examined, evaluating the model's performance using various metrics such as MAE, RMSE, and MAPE. The results, as shown in the table below, indicate that 10 time-lag steps offered the best balance between predictive accuracy and computational efficiency, with no significant performance gains observed with longer sequences. The data were reshaped into 3D tensors, where each sample consisted of a sequence of time steps. In Table 1 the results of the timed step examination for the optimal selection of the time step number are presented.

Table 1. Sequence padding examination.

Time-Lag Steps	MAE (MWh)	RMSE (MWh)	MAPE (%)
5	95.12	110.24	19.2
10	88.72	102.09	17.5
15	89.45	104.58	18.0
20	90.15	106.12	18.3

3.3. Model Development

Main Model with Attention Mechanism

The primary model used in the GREENIA framework was designed to capture both the temporal dependencies and the complex relationships between meteorological features that influence energy production. The architecture integrates GRUs with an attention mechanism, allowing the model to focus on the most relevant time steps and features for accurate forecasting. In Figure 5 a representation of the main ML model that was utilized for the prediction of the energy production is presented.

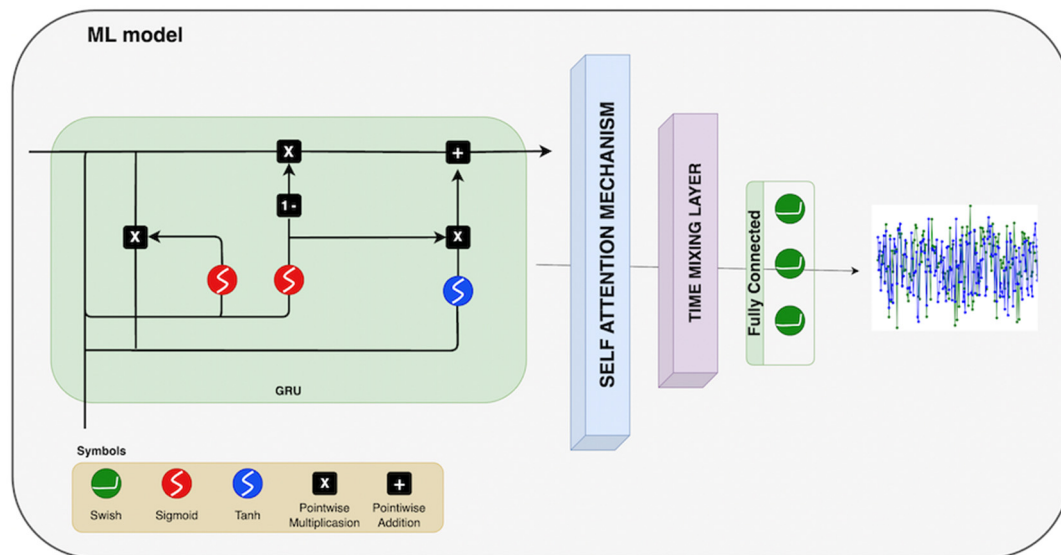


Figure 5. GRU with attention layer (Swish activation function).

- **GRU Layer:** A GRU layer was employed to process the sequential data, leveraging its ability to learn long-term dependencies within the time series. This layer was crucial for capturing the trends and patterns in the meteorological data, such as wind speed and temperature variations, that are critical for accurate energy production forecasts.
- **Attention Layer:** To enhance the model's predictive power, an attention mechanism was applied on top of the GRU layer. The attention layer dynamically assigns weights to different time steps and features, enabling the model to focus on the most relevant portions of the input sequence. This allowed the model to prioritize key moments in the data that have a significant impact on energy production, improving the interpretability and accuracy of the predictions.
- **Weights:** The output of the GRU layer, along with the attention weights, was processed through time-mixing weights—learnable parameters that adapt the importance of different time steps and features. These weights help the model effectively handle varying temporal patterns and feature interactions, enhancing its flexibility in dealing with different types of meteorological sequences.
- **Fully Connected Layers with Swish Activation:** The processed sequence data, influenced by the attention layer, was passed through several fully connected layers, each followed by the Swish activation function. This architecture allowed the model to capture nonlinear interactions between features, further refining its predictive capabilities. The Swish activation function was chosen for its ability to improve model performance through smoother gradient flow during backpropagation.
- **Output Layer:** The final fully connected layer condensed the information into a single output, representing the predicted energy production for the given time period. The model was designed to forecast energy production across different regions and energy types, ensuring accurate and reliable predictions that could be used to guide strategic renewable energy deployment.

3.4. Model Training

The main model was trained using the prepared sequences of meteorological data and corresponding energy outputs.

- **Loss Function:** The mean squared error (MSE) loss function was utilized to quantify the error between the predicted and actual energy production values. This choice was driven by the need to penalize large errors more heavily, which is crucial in the context of energy production forecasting.
- **Optimizer:** The Adam optimizer was chosen for its efficiency in handling sparse gradients and its adaptability to the complex, non-convex optimization landscape typical of deep learning models.
- **Training:** The model was trained for 2000 epochs, with the optimizer iteratively adjusting the model weights to minimize the MSE loss. The training process was carefully monitored to prevent overfitting by utilizing dropout method (dropout = 0.3), and a batch size equal to 256.

3.5. Model Optimization

3.5.1. Jaya Optimization Algorithm

To enhance the model's predictions, the Jaya optimization algorithm was implemented. This algorithm is designed to find the optimal input feature set that maximizes the model's output. The Jaya algorithm iteratively refined the input feature set for 3000 iterations, each time adjusting the inputs to identify the feature combination that yielded the highest energy production predictions. This optimization process was essential for identifying the best possible conditions for maximizing energy output.

3.5.2. Genetic Algorithm (GA) as a Comparative Method

While the Jaya algorithm optimized the input features, a genetic algorithm (GA) was employed to fine-tune the hyperparameters of the model. The GA explored the hyperparameter space by evolving a population of candidate solutions, using crossover and mutation operations to discover the most effective hyperparameter settings.

3.6. SHAP Analysis for Model Interpretability

3.6.1. SHAP Value Calculation

To interpret the model's predictions, Shapley additive explanation (SHAP) values were computed. These values quantify the contribution of each input feature to the model's output, offering insights into which meteorological factors most significantly impact energy production.

- **Kernel Explainer:** The SHAP Kernel Explainer was utilized to approximate SHAP values for the TimeMix model, providing a detailed breakdown of feature importance.
- **SHAP Value Interpretation:** The computed SHAP values identified the most influential features, such as temperature and wind speed, enabling a deeper understanding of the factors driving energy production predictions.

3.6.2. SHAP Analysis Results

The SHAP analysis revealed the most influential meteorological factors for energy production. These factors were then used in the strategic planning phase.

3.7. Strategic Planning for Renewable Energy Deployment

Generating Strategic Insights

The strategic planning process involved synthesizing the results from the optimization and SHAP analysis:

- **GPT-3.5-Turbo-Instruct Integration:** To generate comprehensive strategic recommendations, the GPT-3.5-turbo-instruct model from OpenAI was employed. This model was fed the optimized feature sets and SHAP analysis results and was tasked with

generating a strategic plan for the deployment of renewable energy sources (RESs) across Greece.

- **Strategic Recommendations:** The model generated detailed explanations on how the optimized meteorological features influenced energy production. These explanations were compiled into a strategic report that provided actionable insights on the optimal placement of RESs, such as wind farms and solar panels, based on the geographical and meteorological conditions across different regions in Greece.

The strategic plan was compiled into a comprehensive report that outlines practical guidance for the deployment of RESs across Greece. This plan emphasizes maximizing energy production while considering environmental and social factors.

3.8. Deployment and Documentation

The entire implementation code was uploaded to a GitHub repository [30], making it accessible for public use and further research. The repository includes the following:

- **Source Code Directory:** This includes the Python scripts for data preprocessing, model definition, training, forecasting, optimization, SHAP analysis, and strategic planning.
- **Models Directory:** This includes the trained model, scalers, and SHAP explainer.
- **Outputs Directory:** This contains generated outputs, such as prediction and forecasting results, SHap analysis results, as well as the strategic plan.
- **README.md:** This provides an overview of the project, instructions for use, and a summary of the key findings.
- **.gitignore:** This ensures that unnecessary files, such as large datasets and environment-specific files, are excluded from the repository.
- **Requirements.txt:** This lists all of the dependencies required to run the project, ensuring that others can replicate the environment easily.

In Table 2 the pseudocode that was utilized for the implementation of GREENIA is presented.

Table 2. GREENIA full implementation pseudocode.

GREENIA PSEUDOCODE
Data Pre-process <i># Load and preprocess meteorological data from various locations</i> for each file in raw_data_files: df = read_excel(file, skiprows=9) rename_columns(df, location_name) append_to_main_dataframe(df) <i># Merge all dataframes on common columns (Date, Year, Month, Day, Hour)</i> merged_df = merge_all_dataframes_on_date(main_dataframe) <i># Create 'Date' column and set it as the index</i> merged_df['Date'] = concatenate_year_month_day_hour(merged_df) merged_df.set_index('Date') <i># Resample data to daily frequency</i> resampled_df = resample_dataframe(merged_df, 'D') <i># Handle missing values</i> for each column in resampled_df: if column has more than 65 NaN values: drop_column(column) else: impute_nan_with_mean_of_neighbors(column) <i># Prepare data for model training</i> X, y = split_features_and_target(resampled_df, target_column='Energy (MWh)') X_scaled, y_scaled = scale_features(X, y) X_padded = pad_sequences_to_length(X_scaled, target_length=50)

Table 2. Cont.

GREENIA PSEUDOCODE
Main Model <i>#Define the main model</i> class GRUWithAttention: def __init__(self, input_dim, hidden_dim, output_dim, seq_len): <i># Define GRU layer</i> self.gru = GRULayer(input_dim, hidden_dim) <i># Define attention layer</i> self.attention = AttentionLayer(hidden_dim, num_heads=4) <i># Define fully connected output layer</i> self.fc = FullyConnectedLayer(hidden_dim, output_dim) <i># Define time-mixing weights</i> self.time_mixing_weights = initialize_time_mixing_weights(seq_len, hidden_dim) self.activation = SwishActivation() def forward(self, x): h = self.gru(x) <i># Process through GRU</i> h_att = self.attention(h) <i># Apply attention mechanism</i> h_mixed = apply_time_mixing_weights(h_att, self.time_mixing_weights) <i># Apply time-mixing weights</i> output = self.fc(h_mixed[:, -1, :]) <i># Final output from the last timestep</i> return output
Training <i># Initialize model, loss function, and optimizer</i> model = TimeMixWithAttention(input_dim, hidden_dim=256, output_dim=1, seq_len=10) criterion = MeanSquaredErrorLoss() optimizer = AdamOptimizer(model.parameters(), learning_rate=0.0008) <i># Train model</i> for epoch in range(num_epochs=2000): model.train() outputs = model(X_train_tensor) loss = criterion(outputs, y_train_tensor) backpropagate_and_update_weights(optimizer, loss) log_training_loss(epoch, loss)
Evaluation <i># Evaluate the trained model on the test set</i> model.eval() test_outputs = model(X_test_tensor) test_loss = criterion(test_outputs, y_test_tensor) y_test_orig = inverse_transform(y_test_tensor, scaler_y) test_outputs_orig = inverse_transform(test_outputs, scaler_y) <i># Calculate evaluation metrics</i> mae = calculate_mae(y_test_orig, test_outputs_orig) rmse = calculate_rmse(y_test_orig, test_outputs_orig) mape = calculate_mape(y_test_orig, test_outputs_orig) <i># Plot actual vs forecasted values</i> plot_actual_vs_forecasted(y_test_orig, test_outputs_orig)
Optimization <i># Define the objective function to maximize energy production</i> def objective_function(individual): input_tensor = reshape_and_convert_to_tensor(individual) forecasted_value = model(input_tensor) return -forecasted_value <i># Negative because we minimize</i> <i># Run Jaya optimization for each day</i> for day in range(7): best_individual = jaya_algorithm(objective_function, X_test_tensor[day], bounds) optimize_and_store_results(best_individual, day)

Table 2. Cont.

GREENIA PSEUDOCODE
Explainable Artificial Intelligence (XAI) <i># Define function to get model predictions for SHAP</i> def model_predict_3d(flattened_data): reshaped_data = reshape_to_3d(flattened_data) return model(reshaped_data).detach().numpy() <i># Perform SHAP analysis</i> explainer = shap.KernelExplainer(model_predict_3d, X_test_flat[:30]) shap_values = explainer.shap_values(X_test_flat) shap_table = create_shap_table(shap_values, feature_names) <i># Save SHAP results</i> save_shap_explainer_and_values(explainer, shap_values, save_dir)
Strategic Planning (Natural Language Explanations) <i># Generate strategic insights using GPT-3.5-turbo-instruct</i> for day, best_individual in enumerate(best_individuals): prompt_text = create_prompt(shap_summary, best_individual) explanation = gpt_3_5_turbo_instruct(prompt_text) append_to_strategic_recommendations(explanation) <i># Compile and save the strategic plan</i> strategic_plan = compile_strategic_recommendations() save_strategic_plan_to_file(strategic_plan, save_dir)

4. Results

This section presents the results of the GREENIA strategy, emphasizing the extensive data preparation, prediction, and optimization techniques utilized to forecast and enhance renewable energy output with advanced ML models. The study was performed using a systematic methodology that included data preparation, model training, optimization, and interpretability methodologies to improve forecast accuracy and generate actionable insights for strategic renewable energy development.

4.1. Data Overview and Preparation

After completing the preprocessing steps, the resulting data frame consists of multiple meteorological variables from different weather stations across Greece. The data frame includes columns representing various meteorological attributes such as wind direction, wind speed, temperature, and relative humidity for each station. Additionally, the data frame incorporates a column for the target variable, which is the daily energy production measured in megawatt hours (MWh). A description of the dataset that was utilized for the implementation of the approach is presented as follows:

- The date represents the timestamp for each observation, initially stored in nanoseconds since epoch format.
- Wind direction, maximum wind speed, wind speed, temperature, and relative humidity data were collected from 10 meteorological stations across Greece. These stations were selected based on their geographical diversity, ensuring coverage of regions with high renewable energy potential, as well as data availability and reliability. The selected stations are strategically located in areas with significant solar and wind energy prospects, providing the necessary meteorological input for accurate energy production forecasting.
- Energy defined as the target variable representing the amount of energy produced on each respective day.

After preprocessing, the dataset consisted of 1009 rows, each representing daily observations of weather conditions and the corresponding energy production. The data spanned from the year 2015 to 2018, with daily observations, and included meteorological features from different locations, all of which were converted to numeric format and processed to

remove missing values or filled using the mean of the nearest neighbors. Finally, the daily energy production in MWh was used as the primary variable for prediction.

4.2. Model Training and Evaluation

The training and evaluation of the main model were conducted on a dataset consisting of meteorological features and corresponding energy production values. The results from the training process, alongside the final test performance, are critical in understanding the model's predictive capabilities and its potential limitations.

The training process of the model spanned 2000 epochs, during which the loss function was monitored and minimized using the Adam optimizer. The loss, represented by the mean squared error (MSE), started at a relatively higher value and showed a consistent decline as the training progressed. Specifically, in Figure 6 the training versus validation loss over the training of the model is presented. The training and validation loss curves demonstrate the model's effective learning and generalization over time. Initially, both the training and validation losses are high, indicating that the model starts with significant prediction errors. However, within the first 20 epochs, there is a rapid decrease in both losses, reflecting the model's quick improvement in learning patterns from the data. As training progresses, both losses stabilize at a low value, near 0.02–0.03, suggesting that the model is minimizing errors effectively. The close alignment between the training and validation losses throughout the training process indicates good generalization with minimal signs of overfitting. Small fluctuations observed after epoch 400 are typical during the fine-tuning phase and do not impact the overall performance significantly. These trends suggest that the model is both accurate and robust, performing well on both the training and validation sets.

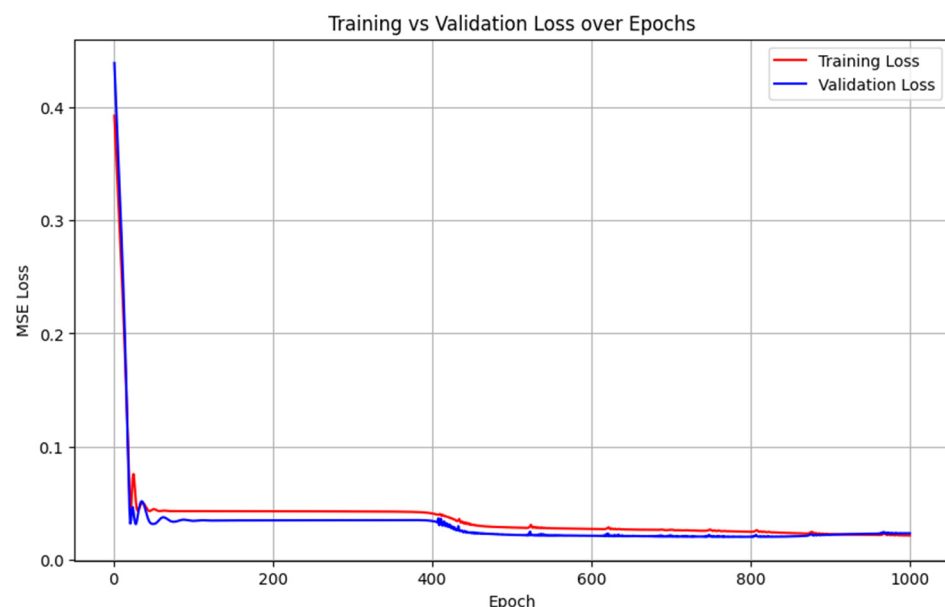


Figure 6. Training vs validation loss curve.

To comprehensively evaluate the model's performance, several metrics were calculated on the test data, including MAE, RMSE, and MAPE. These metrics provide insights into the model's predictive accuracy and the nature of the errors. In Table 3 the evaluation results for the testing phase are presented.

Table 3. Evaluated metrics for the main prediction model.

RMSE	MAE	MAPE
102.09	88.72	17.5%

The MAE was calculated to be approximately 88.72 MWh. This metric provides an average magnitude of the errors in the predictions, without considering their direction. The RMSE was found to be 102.09 MWh. This metric is more sensitive to larger errors due to the squaring of differences before averaging. The higher RMSE compared with MAE indicates that there are some larger deviations in the model's predictions, which could be due to outliers or to periods where the model fails to capture sudden changes in energy production. The MAPE was 17.5%, which is especially useful for understanding the model's performance relative to the scale of the data.

The model was able to predict the output but lacked accuracy in the prediction of abrupt changes and outliers. The comparison of predicted versus actual energy production values, as visualized in the provided graph, highlights several key points, as follows:

- **Alignment and Divergence:** While the predicted values generally align with the actual energy production trends, there are clear instances of both underestimation and overestimation. These discrepancies could be attributed to the model's sensitivity to specific meteorological conditions.
- **Pattern Recognition:** The model appears to capture the broader patterns of energy production, likely driven by recurring meteorological conditions. However, the presence of significant errors in certain periods suggests that the model might not fully account for all relevant factors influencing energy production, such as sudden weather changes or other external variables.

The overall performance of the main model indicates its potential for energy production prediction (Figure 7). However, the significant errors observed, as indicated by the MAE and RMSE, highlight the need for further refinement. The MAPE suggests that the model performed well on average, but the higher absolute error metrics reveal areas where the model struggles, particularly in capturing extreme variations or outliers in the data.

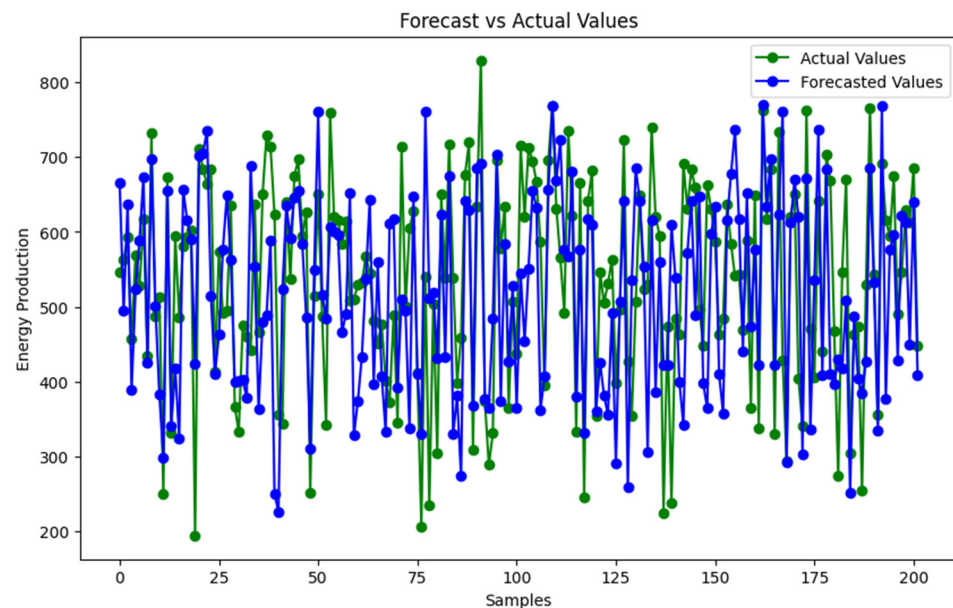


Figure 7. Main model results.

4.3. Forecasting Performance

The forecasting capabilities of the GREENIA framework were evaluated over a 30-day period, leveraging the XGBoost-enhanced GRU model to predict energy production. Figure 8 illustrates the comparison between the forecasted energy production values and the actual observed values across these 30 days.

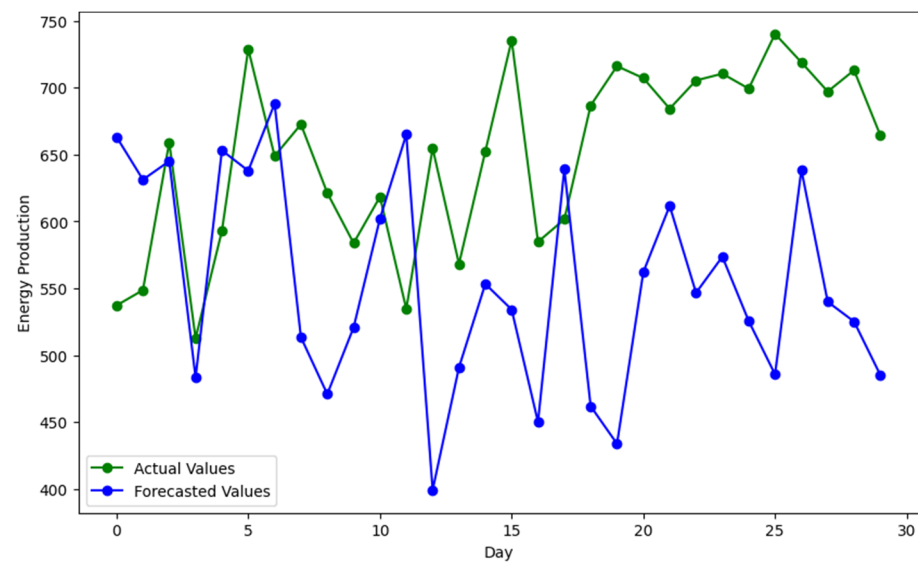


Figure 8. Daily forecasting through AR mechanism.

The green line in Figure 8 represents the actual energy production values, while the blue line indicates the forecasted values. The model's forecasts demonstrate a significant gap between forecasted and actual values, particularly in the second half of the forecasted period, while in the first five days the model is able to forecast close to the actual values. This divergence suggests challenges in capturing the full complexity of energy production dynamics over a longer forecasting window. However, the model is able to forecast the energy production values for the first five days, indicating the model's ability in the short-term horizon. A key observation is the consistent underestimation of energy production by the model, as shown by the blue line frequently trailing below the green line. This trend indicates that, while the model is capable of capturing the general trend of energy production, it struggles to accurately predict higher peaks. This issue may be attributed to the model's sensitivity to certain meteorological variables. Despite these challenges, GREENIA provides valuable insights into energy production forecasting. The overall trends captured by the model offer a useful foundation for strategic energy management, even as the discrepancies highlight areas for future improvement. Accurate medium-term forecasting remains critical for optimizing resource allocation and maintaining grid stability in renewable energy systems.

4.4. Optimization Performance

The optimization phase of the GREENIA framework utilized the Jaya optimization algorithm to enhance the energy production forecasts over a 30-day period. This subsection provides a comprehensive analysis of the quantitative impact of the optimization process, supported by the results shown in Figure 9. The results of the optimization phase demonstrated a significant improvement in energy production forecasts. The pre-optimization forecasted values, depicted by the blue dashed line in Figure 9, were consistently improved upon by the post-optimization values, represented by the orange dashed line. Notably, the light blue bars at the bottom of the graph illustrate the magnitude of improvement for each day.

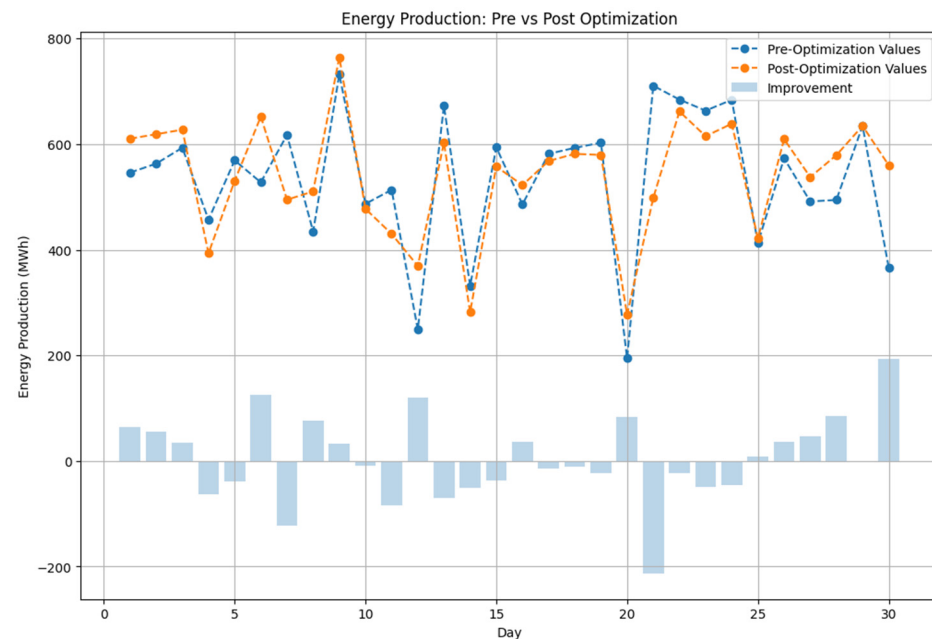


Figure 9. Optimization results for the forecasted energy.

The cumulative effect of the optimization process was significant. The total improvement in forecasted energy production over the 30-day period amounted to 145.57 MWh, highlighting the substantial impact of the optimization process on overall energy output. On average, the optimization process yielded an improvement of 61.60 MWh per day. This consistent daily enhancement underscores the effectiveness of the Jaya optimization algorithm in refining the input features to achieve better energy production forecasts. Although the magnitude of improvements varied across the 30-day period, the optimization consistently improved the forecasted values. On some days, the gains were marginal, suggesting that the initial predictions were already close to optimal. On other days, the improvements were substantial, indicating the optimization's potential to correct less accurate forecasts effectively.

The success of the Jaya optimization algorithm in enhancing energy production forecasts can be attributed to several key factors, as follows:

- **Objective Function:** The objective function was crucial in the optimization process, aiming to maximize the energy production forecast. By minimizing the negative forecasted value, the algorithm effectively manipulated the solution towards optimal conditions that enhanced energy output.
- **Algorithmic Flexibility:** The Jaya algorithm's lack of dependency on specific parameters, such as population size or crossover rate, allowed it to adapt effectively to the problem at hand, focusing solely on improving the objective function.
- **Convergence and Stability:** For 3000 iterations for each day, the algorithm demonstrated stable convergence toward better solutions, as indicated by the overall trend of improvements.

The comparison between pre- and post-optimization results, summarized as follows, underscores the significant value of integrating optimization algorithms within the GREE-NIA framework:

- **Baseline vs. Optimized Forecasts:** Before optimization, the model's forecasts were generally accurate in capturing overall trends but often underestimated energy production, particularly on days with higher variability. The optimization process significantly improved the model's ability to capture these fluctuations, resulting in forecasts that more closely aligned with actual energy production levels.

- **Implications for Strategic Energy Planning:** The improvements achieved through optimization have direct implications for strategic planning in renewable energy deployment. By providing more accurate forecasts, the optimized model allows for better resource allocation and operational planning, enhancing the efficiency and reliability of energy production systems.

Figure 9 illustrates the crucial role of optimization in the GREENIA framework. The consistent improvements observed across the 30-day period validate the effectiveness of the Jaya optimization algorithm in refining energy production forecasts, ultimately contributing to the framework's goal of optimizing renewable energy production. The total improvement of 145.57 MWh, with an average daily improvement of 61.60 MWh and a daily absolute percentage improvement of 13.50%, emphasizes the substantial impact of this optimization process on enhancing energy output, which is critical for maximizing the efficiency of renewable energy systems. Out of the 30 days, the optimization led to positive improvements on 14 days, while 16 days exhibited a decline in forecast accuracy. This resulted in an improvement ratio of 46.67%, indicating that the optimization was beneficial for a significant portion of the period. In Table 4 the quantitative analysis of the optimization component is presented.

Table 4. Optimization phase quantitative analysis.

Metric	Value
Total improvement	145.57 MWh
Mean absolute improvement	61.60 MWh
Mean absolute percentage improvement	13.50%
Positive improvements	14
Improvement ratio	46.67%

4.5. Explainable AI (XAI)

The SHAP table (Table 5) provides valuable insights into the relative importance of various meteorological features in predicting energy production. The features with the highest impact include relative humidity in Kozani (0.0546), maximum 3-h wind speed in Kozani (0.0441), and temperature in Kozani (0.0390). These features stand out as the most significant contributors to the model's predictions, indicating that Kozani's local climate conditions, particularly humidity and wind speed, play a critical role in energy production. The prominence of these features suggests that changes in humidity and wind conditions in this region could lead to substantial variability in energy output, making them key factors to monitor for optimizing renewable energy sources.

Table 5. SHAP values for the input features of the dataset.

Feature	SHAP Value
Temperature (°C)—Kalamata	0.06462843
Relative humidity (%)—Kozani	0.05171667
Temperature (°C)—Kozani	0.04919851
Maximum 3-h wind speed (knots)—Kozani	0.04499661
Relative humidity (%)—Kalamata	0.03563375
Temperature (°C)—Argos	0.03070799
Temperature (°C)—Florina	0.03042352
Temperature (°C)—Ioannina	0.02760436
Temperature (°C)—Kavala	0.02454198
Relative humidity (%)—Florina	0.02201662
Wind direction (°)—Kavala	0.02028739
Relative humidity (%)—Ioannina	0.02000038
Wind speed (knots)—Kalamata	0.01840336

Table 5. Cont.

Feature	SHAP Value
Wind direction (°)—Kozani	0.01649202
Temperature (°C)—Araxos	0.01606719
Temperature (°C)—Agxialos	0.01408858
Relative humidity (%)—Argos	0.01265430
Relative humidity (%)—Agxialos	0.01152817
Wind speed (knots)—Kavala	0.00909824
Wind direction (°)—Kalamata	0.00808502
Wind direction (°)—Argos	0.00747080
Maximum 3-h wind speed (knots)—Argos	0.00710069
Temperature (°C)—Aktio	0.00706860
Maximum 3-h wind speed (knots)—Kavala	0.00673481
Relative humidity (%)—Kavala	0.00671949
Relative humidity (%)—Aktio	0.00597521
Wind direction (°)—Aktio	0.00591145
Wind direction (°)—Florina	0.00578259
Relative humidity (%)—Alexandroupoli	0.00551289
Maximum 3-h wind speed (knots)—Florina	0.00513565
Relative humidity (%)—Araxos	0.00485569
Wind direction (°)—Agxialos	0.00472039
Maximum 3-h wind speed (knots)—Kalamata	0.00388574
Wind direction (°)—Araxos	0.00363804
Maximum 3-h wind speed (knots)—Araxos	0.00341315
Maximum 3-h wind speed (knots)—Agxialos	0.00319952
Wind speed (knots)—Aktio	0.00318487
Temperature (°C)—Alexandroupoli	0.00318217
Wind speed (knots)—Florina	0.00304688
Wind direction (°)—Alexandroupoli	0.00303900
Wind speed (knots)—Argos	0.00295486
Wind speed (knots)—Agxialos	0.00252417
Maximum 3-h wind speed (knots)—Alexandroupoli	0.00241123
Wind speed (knots)—Kozani	0.00215932
Maximum 3-h wind speed (knots)—Aktio	0.00203236
Wind speed (knots)—Araxos	0.00170344
Wind direction (°)—Ioannina	0.00009577

The SHAP analysis provided insights into which meteorological features have the most significant impact on energy production. Below is a detailed explanation of the top features based on their SHAP values:

- **Temperature (°C)—Kalamata:** With the highest SHAP value, the temperature in Kalamata is the most influential feature in the energy production model. This suggests that fluctuations in temperature in this region greatly affect the amount of energy produced.
- **Relative Humidity (%)—Kozani:** Relative humidity in Kozani is the second most impactful feature. High or low humidity levels in this area are key indicators of changes in energy production.
- **Temperature (°C)—Kozani:** Similar to Kalamata, the temperature in Kozani also plays a crucial role in predicting energy output. This indicates the importance of temperature variations in this region.
- **Maximum 3-h Wind Speed (knots)—Kozani:** This feature represents the maximum wind speed recorded over a 3-h period in Kozani. Its high SHAP value highlights wind speed's critical role in driving energy production, particularly in windy regions like Kozani.

- Relative Humidity (%)—Kalamata: In addition to temperature, humidity in Kalamata is also a significant factor, further emphasizing the role of meteorological conditions in this region.
- Temperature (°C)—Argos: Temperature in Argos is another vital feature, indicating that this region's climate heavily influences energy production.
- Temperature (°C)—Florina: The temperature in Florina is crucial, although to a slightly lesser extent than in the previously mentioned areas. It is still a significant predictor in the model.
- Temperature (°C)—Ioannina: Ioannina's temperature also contributes notably to the energy production forecast, reinforcing the importance of temperature as a feature across multiple regions.
- Temperature (°C)—Kavala: Temperature variations in Kavala are another important factor, although with a somewhat lower impact compared with the top-ranking features.
- Relative Humidity (%)—Florina: Similar to Kozani, humidity in Florina is a critical element in the model, affecting the energy output predictions.

These SHAP values collectively indicate that temperature and humidity are the dominant factors influencing energy production, with wind speed and direction also playing significant roles, particularly in regions like Kozani and Kalamata. Understanding these variables' contributions helps in optimizing renewable energy resource deployment and maximizing efficiency based on the local climate conditions of each region.

4.6. Strategic Planning for Renewable Energy Deployment

The strategic planning for renewable energy deployment in Greece, based on the analysis conducted using the GREENIA framework, offers a comprehensive approach to optimizing the placement of renewable energy sources (RESs) across the country. This section details the strategic recommendations derived from the SHAP analysis and the optimization results, focusing on maximizing energy production efficiency and sustainability.

4.6.1. Wind Energy

The SHAP analysis highlighted that wind speed and direction are critical factors influencing wind energy production. The highest SHAP values for these features were found in Kozani, Agxialos, Aktio, and Aleksandroupoli. These locations exhibit optimal conditions for wind energy generation, particularly due to their favorable wind patterns.

- Kozani: Wind turbines should be strategically placed in areas with consistent high wind speeds and aligned with the prevailing wind directions to maximize energy capture.
- Agxialos and Aktio: These coastal regions benefit from strong sea winds, making them ideal for the installation of wind turbines along the shoreline. The coastal wind patterns here can significantly enhance wind energy production.

These recommendations are supported by the optimization results, which showed a significant improvement in energy production when the wind turbines were placed in these regions. Environmental impact assessments and community consultations are crucial to ensure that the deployment of wind turbines is both sustainable and socially acceptable.

4.6.2. Solar Energy

Temperature and humidity were identified as the most influential factors for solar energy production, with the highest SHAP values observed in Kalamata, Kozani, and Argos. These regions, characterized by high temperatures and relatively low to moderate humidity levels, offer optimal conditions for solar energy generation.

- Kalamata: Solar panels should be installed in areas with the highest temperatures and lowest humidity levels to maximize energy output. The region's climate is particularly conducive to solar energy production.

- Kozani and Argos: In these regions, solar installations should focus on locations with consistent high temperatures and moderate humidity, ensuring optimal performance of solar panels.

The strategic plan suggests prioritizing land availability and minimizing shading effects from surrounding structures or natural features when selecting specific sites for solar farms.

4.6.3. Hydro Energy

While not a primary focus of the GREENIA framework, hydro energy potential was also assessed, particularly in regions where wind speed and direction play a role. Kozani and Argos were identified as favorable locations for hydro energy installations, particularly near rivers or other bodies of water with strong currents.

- Kozani: Hydro turbines should be positioned in areas with high wind speeds, which can influence water flow and enhance energy production.
- Argos: The proximity to strong water currents makes this region suitable for hydro energy projects, especially in conjunction with other renewable sources like wind.

Environmental considerations, such as the impact on local ecosystems and water use, must be thoroughly evaluated during the planning phase.

4.6.4. Geothermal Energy

Geothermal energy potential is largely driven by temperature, with Kozani emerging as the most promising location based on the SHAP analysis.

- Kozani: The region's high temperatures suggest significant geothermal activity, making it an ideal site for geothermal power plants. Strategic placement of geothermal wells should focus on areas with accessible geothermal reservoirs.

Geological surveys and environmental impact assessments are essential to confirm the viability of geothermal energy production in Kozani.

4.6.5. Biomass Energy

The SHAP analysis indicated that temperature and humidity are also critical for agricultural biomass energy production, with Kozani and Argos identified as optimal locations.

- Kozani and Argos: Biomass power plants should be located in regions with high temperatures and moderate humidity, which are conducive to the efficient conversion of biomass into energy. The availability of biomass resources in these regions should also be assessed to ensure a sustainable supply chain.

Community engagement and sustainability assessments are necessary to address potential conflicts over land use and resource allocation.

The strategic plan for deploying renewable energy sources across Greece is grounded in the detailed analysis provided by the GREENIA framework. By prioritizing regions with the highest SHAP values for each energy type, the plan ensures that energy production is maximized while maintaining a focus on sustainability and social acceptance.

The plan emphasizes the importance of considering environmental and social impacts when selecting specific locations for RES deployment. Collaboration with local communities, thorough environmental assessments, and adherence to regulatory standards are all critical components of a successful renewable energy strategy.

This strategic plan provides a roadmap for the future development of renewable energy in Greece, contributing to the country's goals of increasing energy independence, reducing carbon emissions, and promoting sustainable growth. The insights generated from the GREENIA framework can be used to guide decision making, optimize resource allocation, and ensure that renewable energy projects are both effective and sustainable.

5. Comparative Analysis of Renewable Energy Deployment Parameters

In the strategic planning for renewable energy deployment, various critical aspects affect the best location of RESs, encompassing geographic, environmental, and social considerations. The GREENIA framework offers a holistic and comprehensive approach to the deployment of RESs by integrating advanced ML techniques, multi-source optimization, and explicability through SHAP analysis. To further validate the effectiveness and uniqueness of the framework, we compare its methodology and results with existing studies on renewable energy optimization. The following comparative analysis highlights the strengths of GREENIA relative to several other approaches in the field.

Chantzis et al. [27] concentrated on enhancing a hybrid renewable energy system for non-interconnected islands in Greece, with a specific emphasis on Amorgos. Their research employed linear programming to reduce the levelized cost of energy (LCOE), using wind, solar, and battery storage to achieve energy independence. This technique was used to overcome the distinct energy difficulties of isolated islands, although it predominantly centered on a singular case study with restricted applicability to other geographic contexts. In contrast, the GREENIA framework goes beyond optimizing hybrid systems by considering a wide array of renewable energy sources, such as wind, solar, geothermal, hydro, and biomass, across mainland Greece. By employing SHAP analysis for transparency and the Jaya algorithm for optimization, GREENIA provides a more comprehensive solution that not only addresses technical aspects but also incorporates social and environmental sustainability, ensuring that renewable energy deployment is well aligned with community needs and long-term ecological goals.

Furthermore, Wu et al. [31] concentrated on enhancing the siting of onshore wind farms, especially addressing wake impacts and topographical considerations. Their application of particle swarm optimization (PSO) was efficacious for optimizing wind farm layouts; however, it is constrained to the technological dimensions of wind energy production. Furthermore, their research focused predominantly on cost efficiency and the reduction of energy losses, neglecting multi-source energy systems and the wider social and environmental implications. Conversely, the GREENIA framework integrates many energy sources and employs a more adaptable optimization method suitable for a wider array of geographic and environmental contexts. Moreover, GREENIA considers community consultation, land availability, and environmental impact assessments, rendering it a more comprehensive solution that harmonizes technological optimization with social and environmental considerations.

Considering distributed energy placement, Gkaidatzis et al. [32] addressed the optimal distributed generation placement (ODGP) problem using uPSO to reduce power losses in distribution networks. While this approach was highly efficient for minimizing energy losses and optimizing distributed energy resource (DER) placement, it was largely focused on technical optimization within the grid and did not take into account social, environmental, or economic factors that might influence the long-term sustainability of renewable energy projects. The GREENIA framework integrates multi-source optimization and addresses technical, social, and environmental considerations. It goes beyond simply reducing energy losses by incorporating community engagement and environmental impact assessments to ensure that the deployment of renewable energy sources is both technically optimal and socially sustainable.

Finally, another main component of the GREENIA framework is the XAI. Rojas et al. [33] explored the use of XAI and packetized energy management (PEM) in industrial energy systems, focusing particularly on optimizing energy loads in industrial processes using networked sensing and software-defined energy networks. Their approach focused on industrial energy management through PEM and software-defined energy networks (SDEN), which primarily aimed at optimizing energy distribution in industrial plants by dynamically managing energy packets and scheduling processes. The implementation integrated AI but its primary goal was real-time energy optimization in industrial settings. The GREENIA framework incorporates XAI through SHAP analysis, but it applies this

technique to strategic decision making for renewable energy deployment at a national or regional scale. This ensures that decisions are transparent not only in industrial settings but also in large-scale energy planning that involves multiple stakeholders, including local communities and policymakers.

6. Limitations and Future Work

Despite the promising results achieved through the GREENIA framework, several limitations must be acknowledged, which also pave the way for future research and improvements. One of the primary limitations of this study is the reliance on meteorological and energy production data from specific regions in Greece. The accuracy and generalizability of the GREENIA framework are contingent upon the availability and quality of these data. In regions with sparse or low-quality data, the predictions and optimization outcomes may be less reliable. Future work should explore methods to enhance data collection, such as integrating remote sensing technologies or using crowdsourced weather data, to improve model performance in data-scarce regions.

While the GREENIA framework provides valuable insights for strategic planning, it has not yet been integrated into real-time energy management systems. The ability to adapt to real-time data and dynamically adjust energy production strategies in response to changing conditions would significantly enhance the framework's utility. Future research should explore the integration of GREENIA with real-time monitoring and control systems, potentially using internet of things (IoT) devices and edge computing to enable rapid, automated decision making.

The optimization and strategic planning processes in GREENIA primarily focus on maximizing energy production. However, they do not fully account for social and environmental factors, such as community opposition to RES placement, land use conflicts, or the potential impact on local ecosystems. Future work should incorporate multi-objective optimization techniques that balance energy production with social and environmental sustainability. Engaging with local communities and stakeholders to ensure that renewable energy projects are both effective and socially acceptable is also a critical area for future research.

While GREENIA's forecasting component is effective in the short term, it may be subject to increased uncertainty when extended to long-term predictions. Changes in climate patterns, economic conditions, and technological advancements could all impact the accuracy of long-term forecasts. Future research could explore ways to incorporate these uncertainties into the model, potentially using probabilistic forecasting methods or scenario analysis to provide a range of possible outcomes rather than a single deterministic forecast.

As AI and machine learning play an increasingly prominent role in energy systems, ethical considerations and policy implications must be addressed. The deployment of AI-driven energy strategies raises questions about transparency, accountability, and fairness, particularly in how decisions impact different communities. Future work should examine the ethical dimensions of AI in energy management, ensuring that the benefits of AI are distributed equitably and that decision-making processes remain transparent and accountable.

In conclusion, while the GREENIA framework represents a significant advancement in the strategic deployment of renewable energy sources, addressing these limitations will be essential to fully realize its potential. By enhancing data quality, improving computational efficiency, and by integrating social, environmental, and ethical considerations, future research can build upon the foundation laid by GREENIA to create even more robust, flexible, and equitable energy systems.

7. Conclusions

The development and deployment of RESs are pivotal in the global transition towards a sustainable future. In this context, the GREENIA framework emerges as a pioneering

solution tailored specifically to optimize the strategic placement and utilization of RESs in Greece. This framework not only integrates advanced machine learning techniques with powerful optimization algorithms but also emphasizes transparency and interpretability, which are crucial for informed decision-making in energy management.

The current study highlights the significant advancements achieved through the GREENIA framework. By employing predictive models, such as GRU with attention mechanisms, and coupling them with robust optimization strategies like the Jaya algorithm, GREENIA delivers marked improvements in forecasting accuracy and optimized energy outputs. These improvements are quantitatively demonstrated through the substantial gains in energy production forecasts, as well as the strategic insights provided for deploying RESs across Greece.

The framework's integration of SHAP analysis further enhances its utility by offering a transparent view of how different meteorological factors influence energy production predictions. This transparency is essential for stakeholders who require a deep understanding of the factors driving energy outputs, ensuring that the decisions made are both data-driven and comprehensible.

Moreover, the strategic planning component of GREENIA goes beyond mere optimization by offering a comprehensive roadmap for RES deployment. It considers critical meteorological variables across diverse geographical regions, ensuring that the placement of energy infrastructures like wind farms and solar panels is done in a manner that maximizes efficiency and sustainability. This level of strategic insight is invaluable for regional planners and policymakers tasked with implementing Greece's renewable energy goals.

Looking forward, the GREENIA framework's potential extends far beyond this study's immediate results. As global challenges like climate change demand increasingly sophisticated solutions, frameworks like GREENIA could play a crucial role in shaping the future of energy systems. Integrating technical innovation with strategic foresight and transparency positions GREENIA as not just a tool for optimization but as a comprehensive framework for sustainable energy management.

Future research and development should focus on enhancing the framework by addressing its current limitations and exploring its applicability in other regions and contexts. This includes improving data quality and availability, incorporating real-time system integration, and expanding the framework's scope to include social, environmental, and ethical considerations. As the framework evolves, it could be adapted to accommodate the complexities of different energy markets, regulatory environments, and technological advancements, further solidifying its role in the global energy transition.

In conclusion, this research is not only able to advance the scientific understanding of renewable energy system optimization through advanced deep learning and evolutionary algorithms but also provides significant social benefits. The GREENIA framework contributes to a more sustainable energy future by optimizing the strategic placement of RESs, leading to enhanced efficiency and energy output. This optimization process can directly support policy making, improve energy independence, and reduce carbon emissions, aligning with the broader societal goals of combating climate change and ensuring a reliable and equitable energy supply. By providing explainable, data-driven recommendations, GREENIA empowers stakeholders, including policymakers, energy planners, and local communities, ensuring that energy solutions are transparent and aligned with community needs.

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